

TOI-4087b

R-2193

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI4087_260626 · created 2026-08-13T23:59:42+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461214.6296 +/- 0.0085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.134 +/- 0.026
Transit depth (Rp/Rs)^2	1.78 +/- 0.69 [%]
Orbital Inclination (inc)	86.71 +/- 1.34
Airmass coefficient 1 (a1)	0.624 +/- 0.0066
Airmass coefficient 2 (a2)	0.287 +/- 0.093
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	None
Optimal Aperture	2.91
Optimal Annulus	13.08
Transit Duration (day)	0.034 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.134 $\pm$ 0.026 vs 0.10754068997360952 (deviation 0.87 σ)
Duration fit vs published	0.034 d vs 0.1242 d (deviation 7.04 σ)
Mid-time O–C	-4539.0 min
Depth SNR	4.4
Residual scatter	1.05 %

Light curve

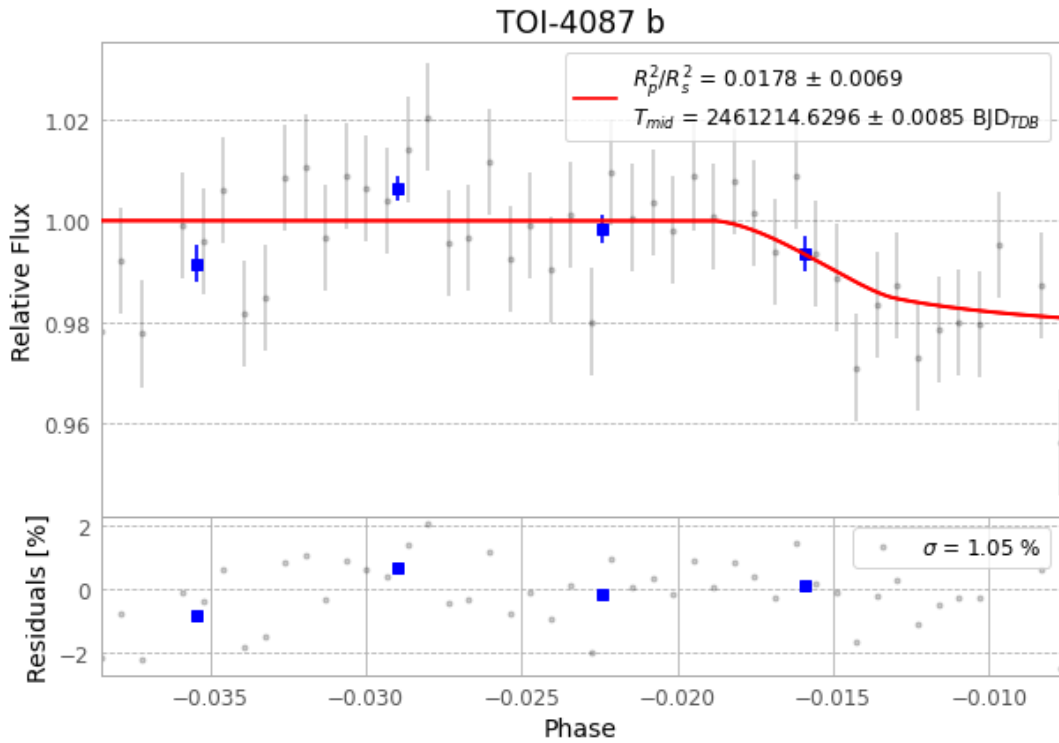


Figure 1 — FinalLightCurve_TOI-4087_b_2026-06-25.png (SHA-256 fcf5daab41102a5d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [340, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d88e1ade8d6c1b96...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

48 FITS frames (32 MB), TOI4087260626042715.FITS → TOI4087260626064815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4087-260626.log (6ca35f4001be...), FinalLightCurve_TOI-4087_b_2026-06-25.png (fcf5daab4110...), FinalLightCurve_TOI-4087_b_2026-06-25.csv (e97fe8f12ea5...)

Compute resources

Wall time 143.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:59Z.